Lab: Extract Data from Aurora MySQL Using AWS Glue Python Shell

Objective

Create a Python Shell AWS Glue job that extracts data from Amazon Aurora MySQL tables, supports full and incremental loads using DynamoDB configuration, and writes extracted data to an S3 landing zone.

Step 1: Understand the MySQL Extraction Script

1. Locate the script named **mysql_extraction.py**.
2. The script uses:
   - `pymysql` to connect to Aurora MySQL.
   - `boto3` to interact with S3, DynamoDB, and Secrets Manager.
3. The job accepts two runtime arguments:
   - `-table_name`
   - `-load`
4. The `load` argument can be:
   - `incremental`
   - Any other value (treated as full load)
5. The script:
   - Fetches RDS credentials from **AWS Secrets Manager**
   - Connects to the `rental_apartments` database
   - Extracts data based on load type
   - Writes output as `data.csv` to:

```
s3://<bucket>/landing-zone/apartment-db/<table_name>/data.
csv
```

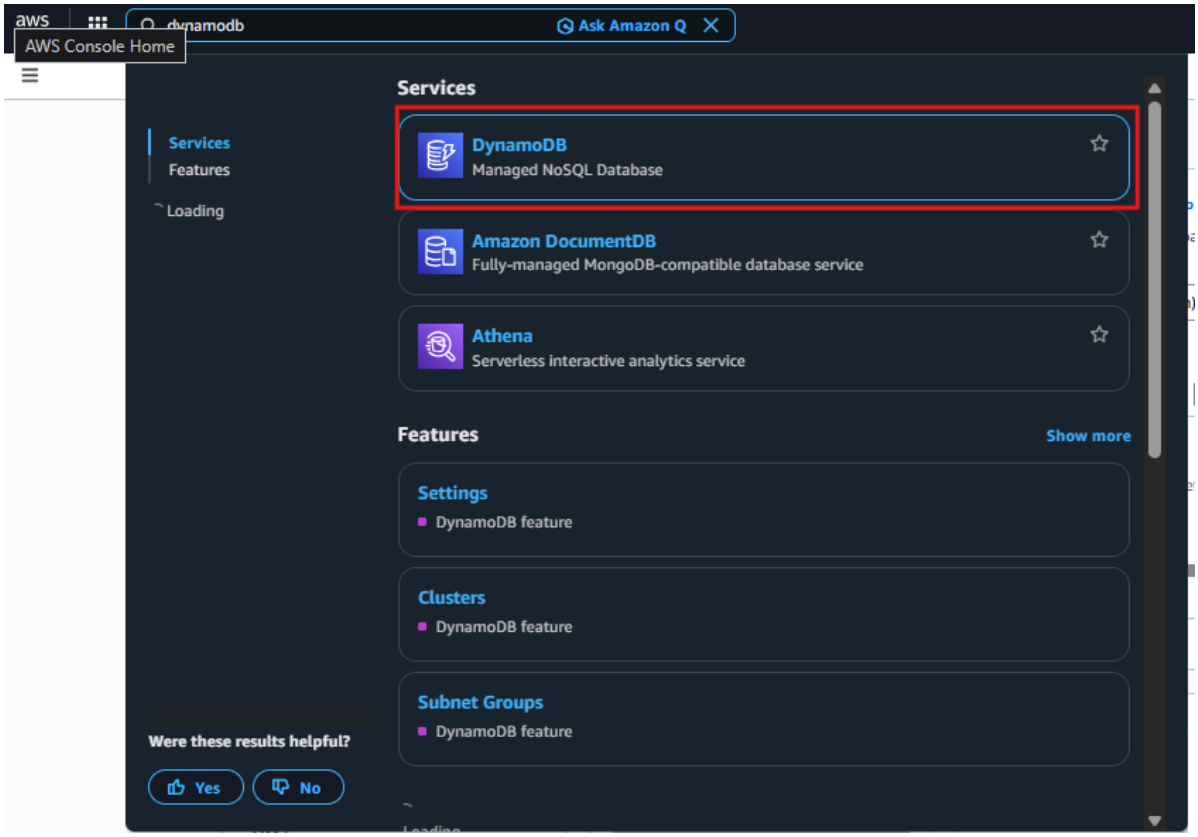
6. For incremental loads:
   - It retrieves configuration from DynamoDB table: `IncrementalLoadConfigurations`
   - Uses:
     - Incremental column (e.g., last_modified_timestamp or viewed_at)
     - Last extracted value
   - Executes query:

```
SELECT * FROM table_name
WHERE incremental_column > last_extracted_value;
```

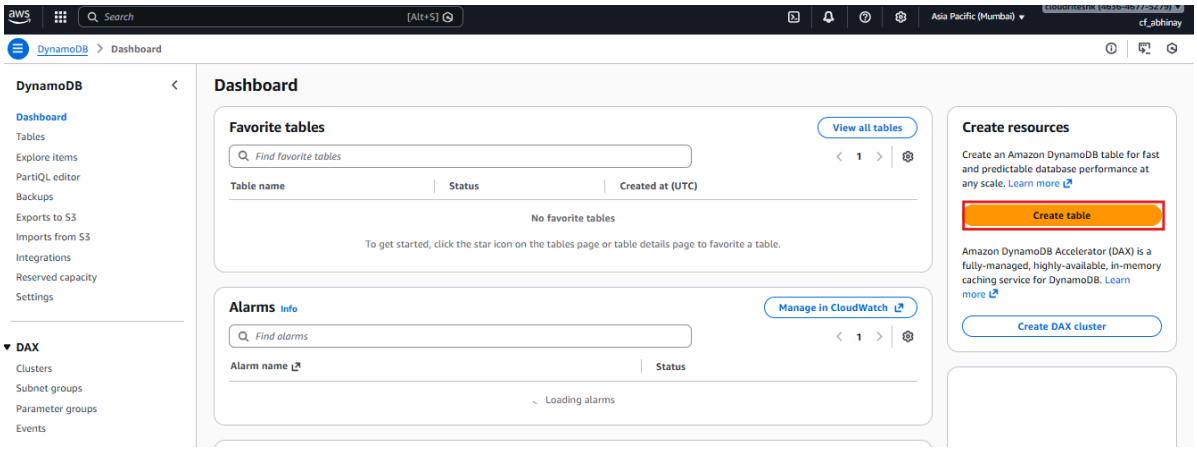
7. After successful extraction, it updates the `last_extracted_value` in DynamoDB.

Step 2: Create DynamoDB Table for Load Configuration

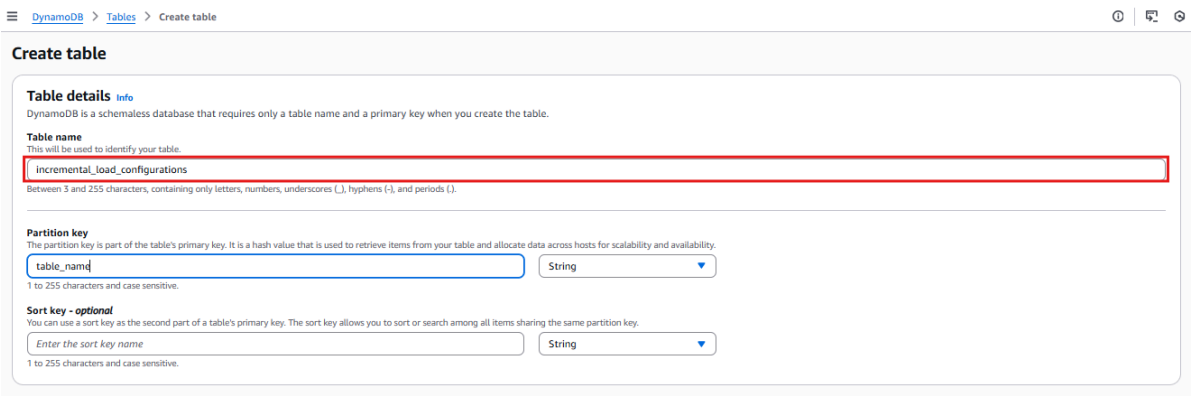
1. Open **AWS Console** and search for **DynamoDB**.



2. Click **Create table**.

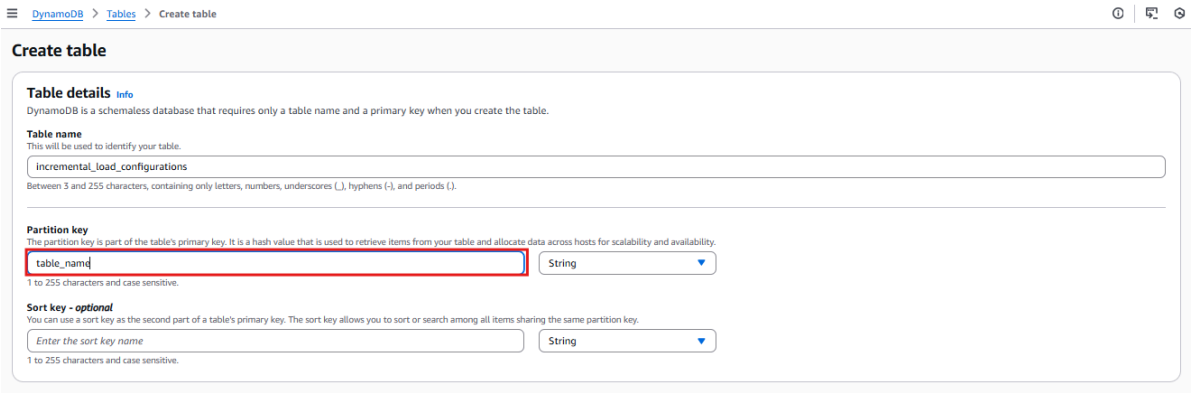


3. Table name: `IncrementalLoadConfigurations`

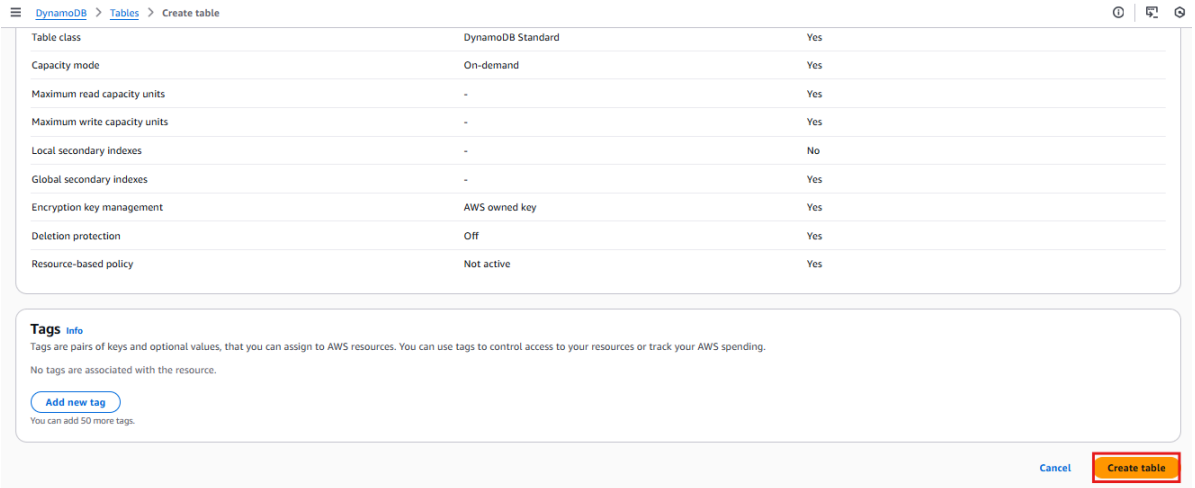


4. Partition key:

   - `table_name`



5. Leave remaining settings as default.
6. Click **Create table**.



Step 3: Run write_to_dynamo.py to Insert Configuration

1. Open the script `write_to_dynamo.py`.
2. This script inserts three configuration records:
   - apartments
     - load column → `last_modified_timestamp`
     - last_extracted_value → NULL
   - apartment_attributes
     - load column → NULL (full load table)
     - last_extracted_value → NULL
   - apartment_viewings
     - load column → `viewed_at`
     - last_extracted_value → NULL
3. Before running the script:
   - Ensure AWS CLI is configured:

   ```
   aws configure
   ```

4. Navigate to the working directory.
5. Run:

   ```
   python3 write_to_dynamo.py
   ```

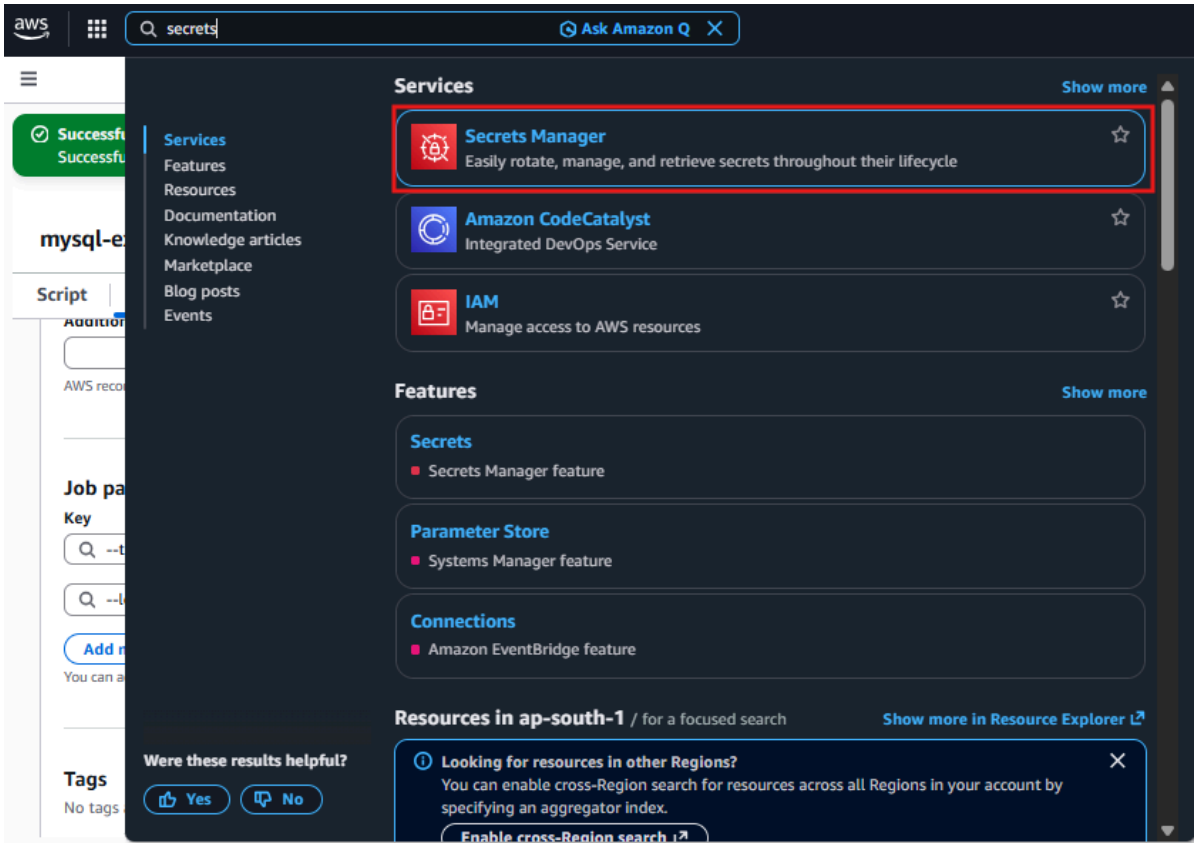
6. Verify entries in DynamoDB using **Explore items** or **PartiQL Editor**:

   ```
   SELECT * FROM incremental_load_configurations;
   ```

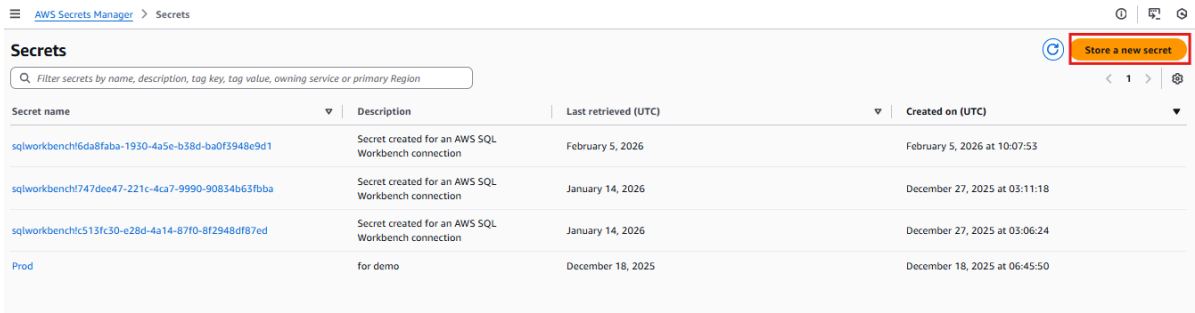
You should see three records with NULL last_extracted_value.

Step 4: Create Secret in AWS Secrets Manager

1. Search for **Secrets Manager** in AWS Console.



2. Click **Store a new secret**.

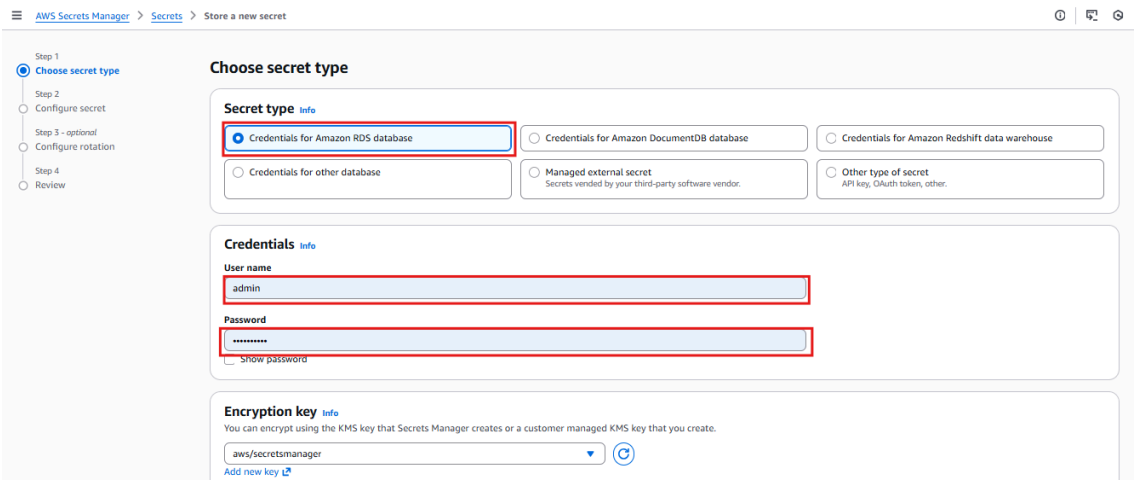


3. Choose:

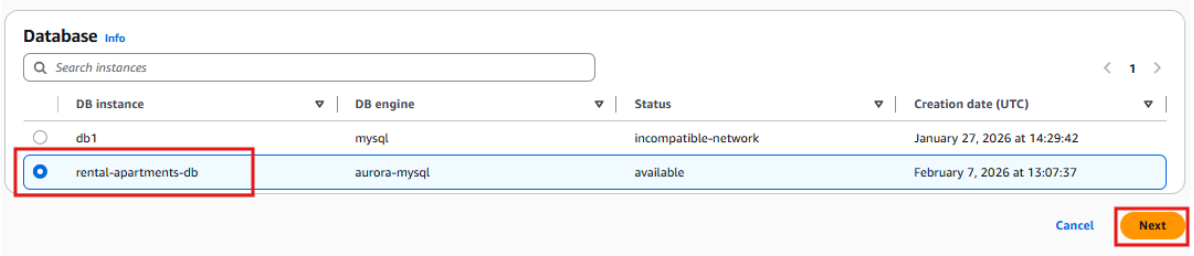
   - Credentials for Amazon RDS database

4. Enter:

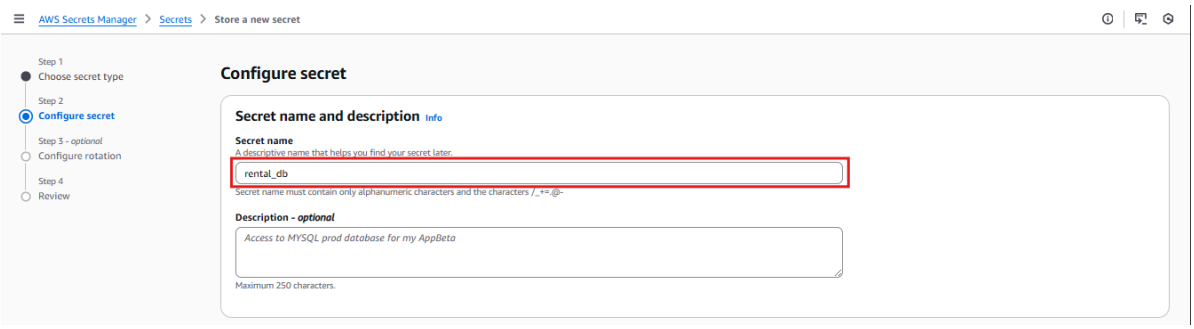
- Username → `admin`
- Password → (Aurora password)



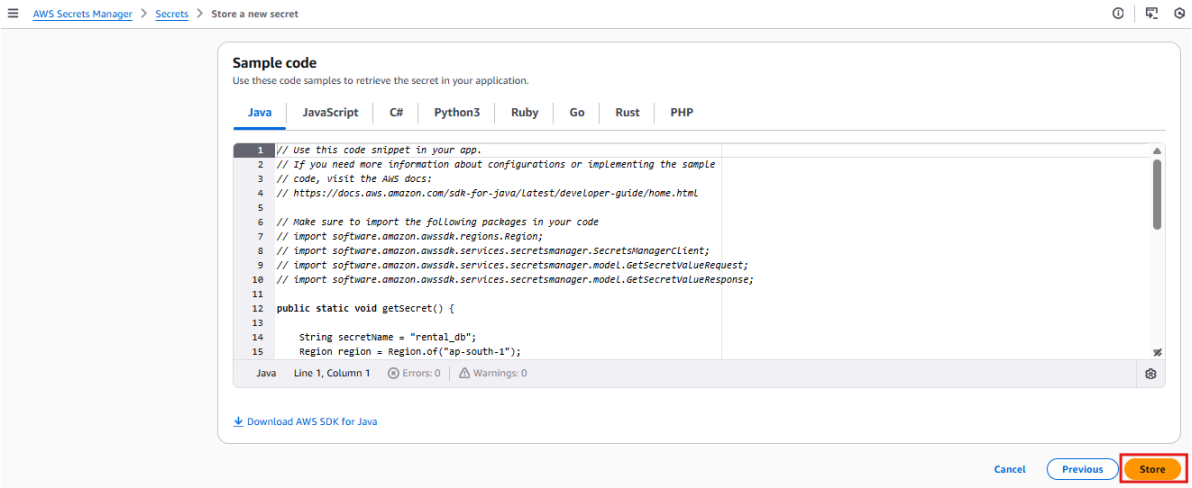
5. Select the Aurora database created earlier.



6. Secret name → `rental_db`



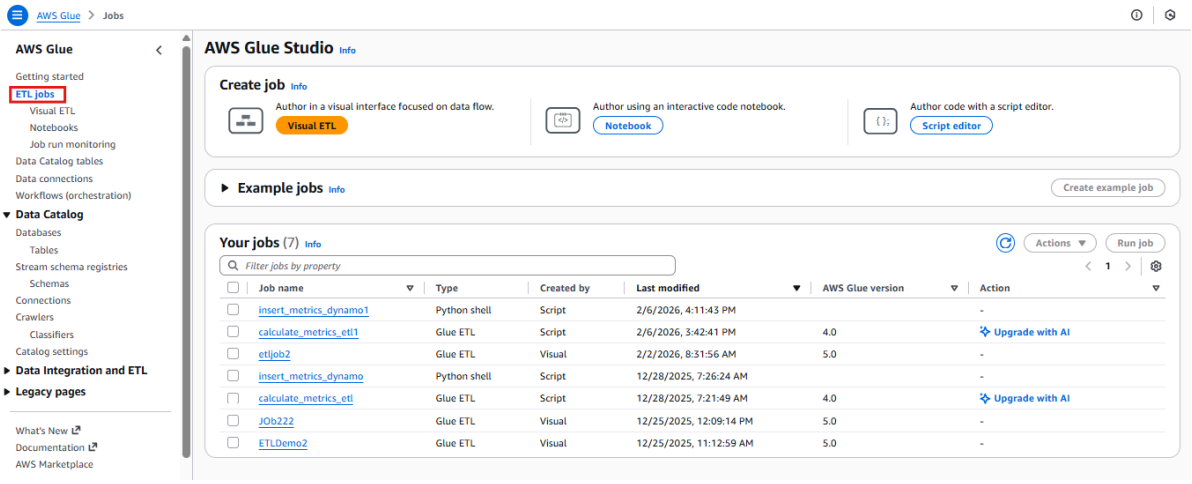
7. Click **Next** → **Next** → **Store**.



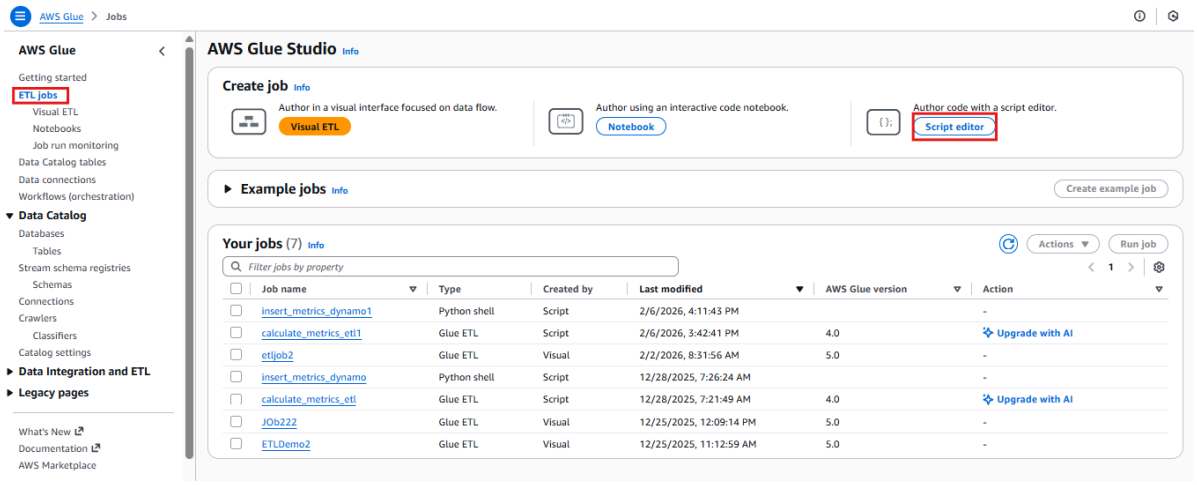
8. Verify by clicking **Retrieve secret value**.

Step 5: Create AWS Glue Python Shell Job

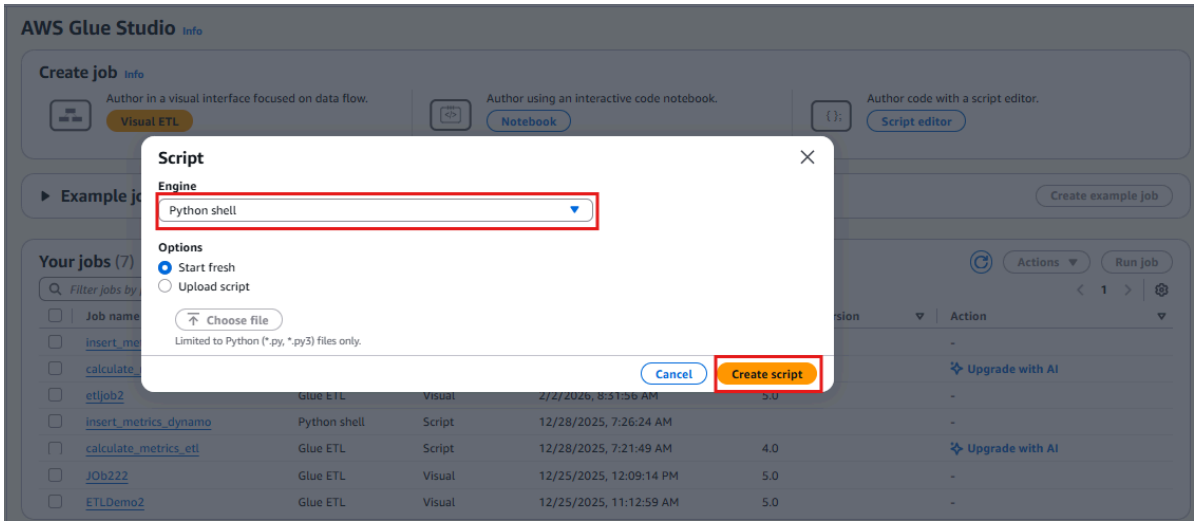
1. Navigate to **AWS Glue** → **ETL Jobs**.



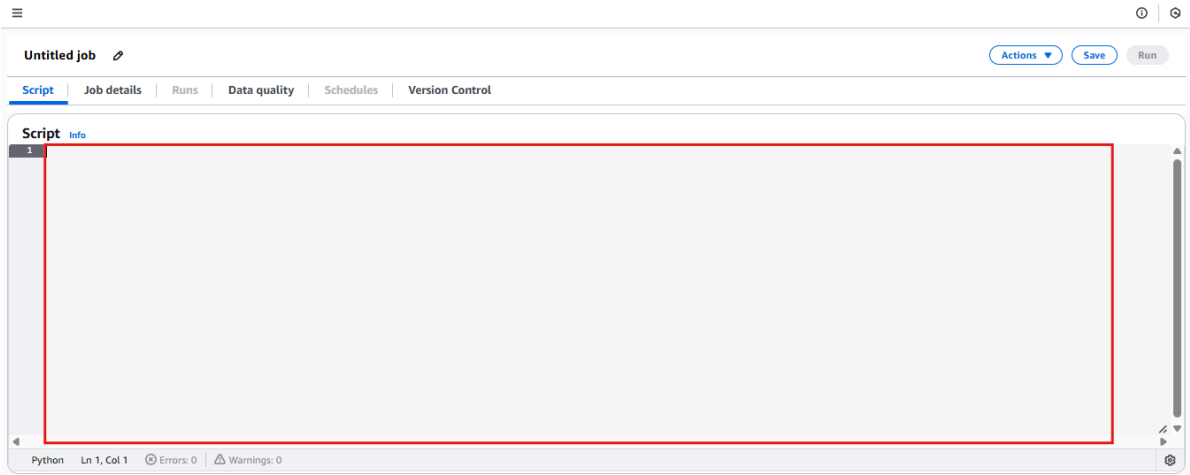
2. Click **Script editor**.



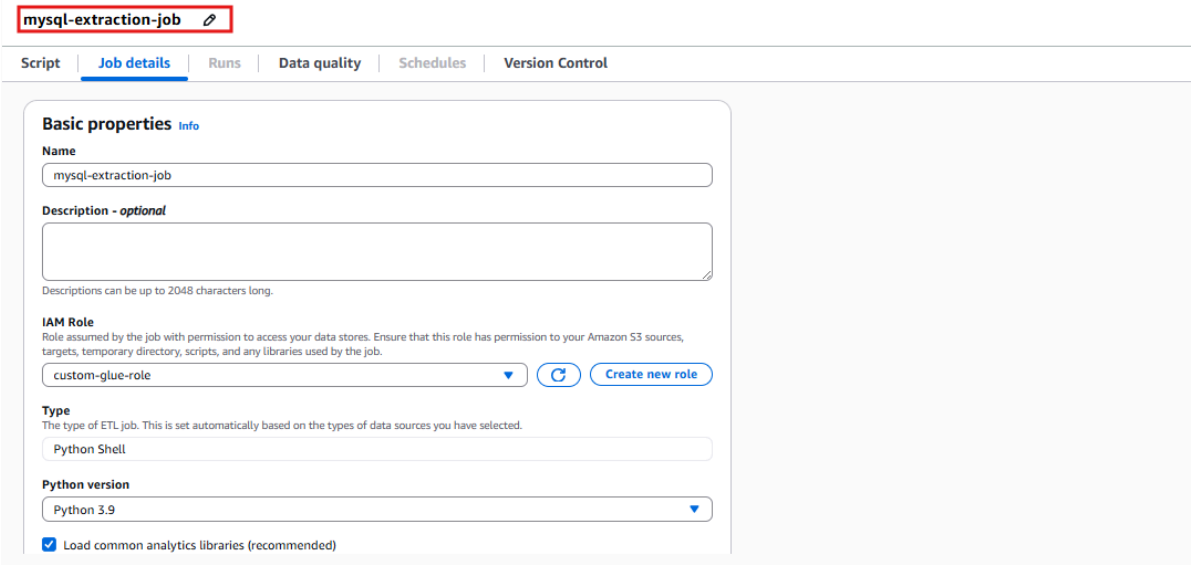
3. Select **Python Shell**.



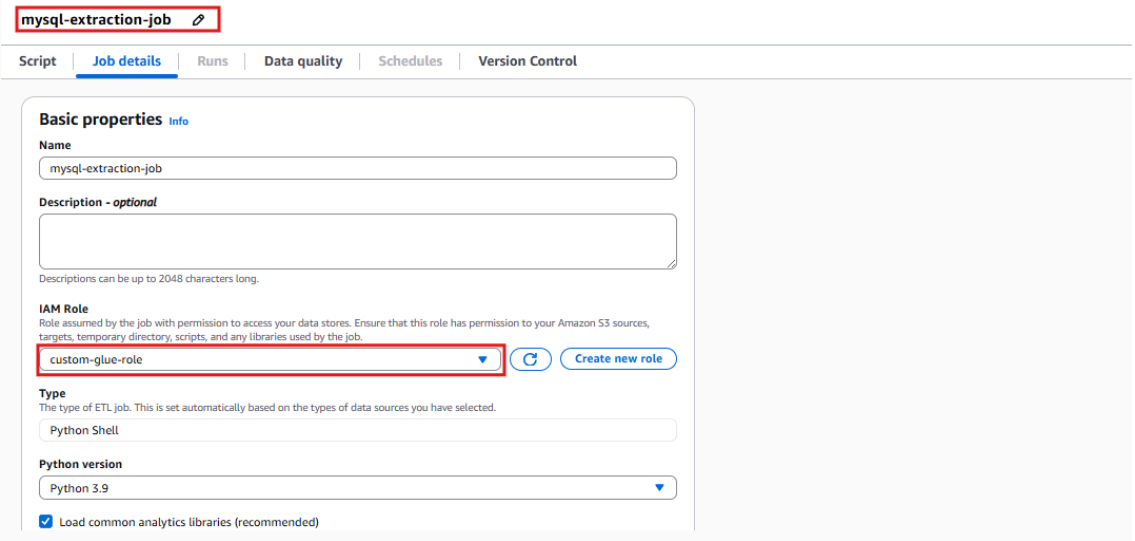
4. Paste contents of `mysql_extraction.py`.



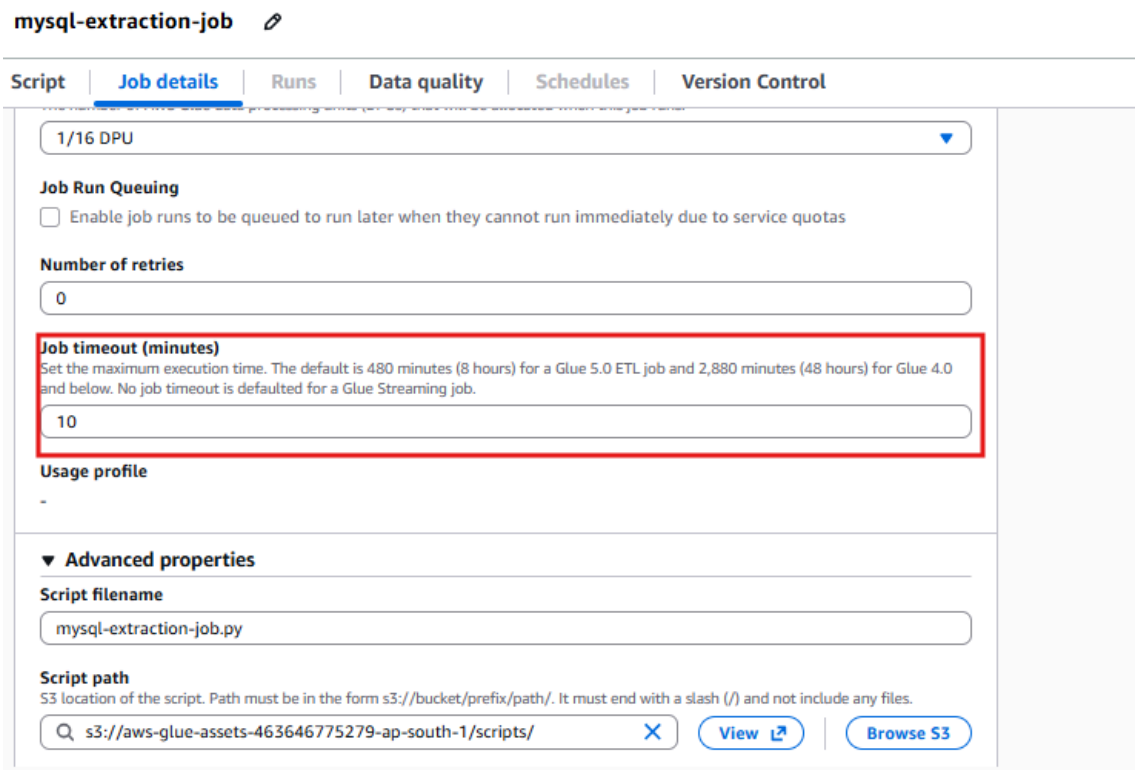
5. Update:
    - S3 bucket name
    - AWS region (if required)
6. Job name → `MySQLExtractionJob`



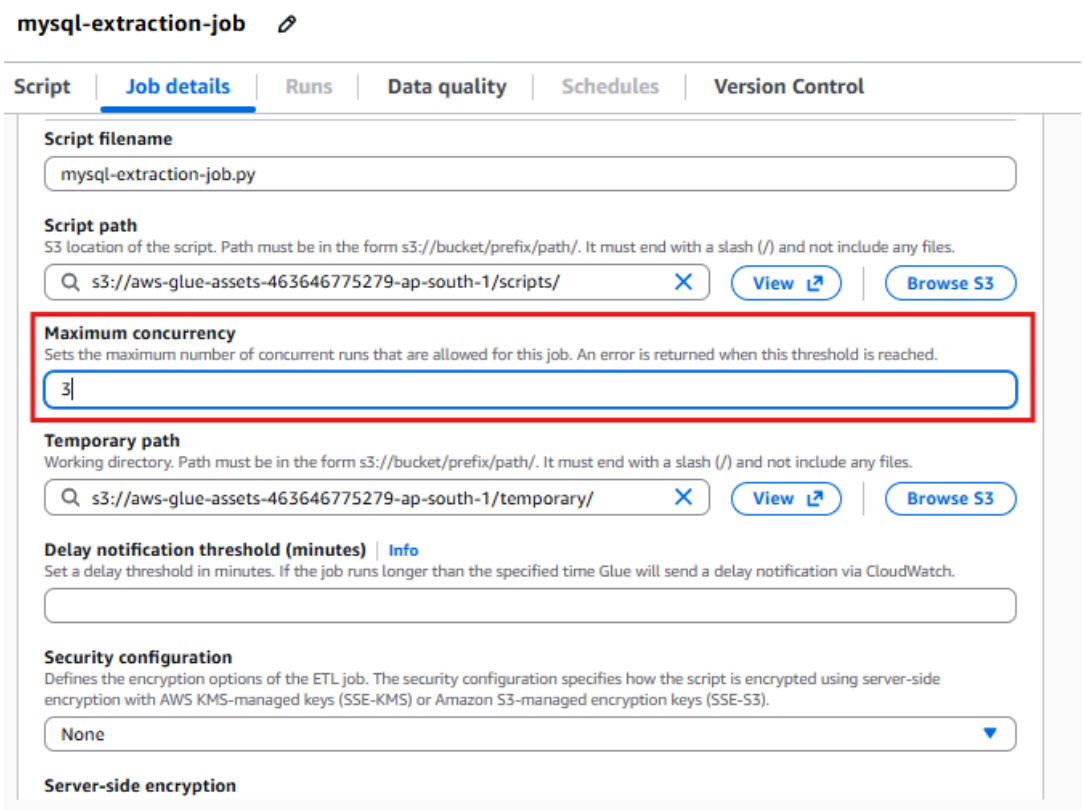
7. In **Job details**:
    - Select IAM role (must have access to S3, DynamoDB, CloudWatch, Glue)



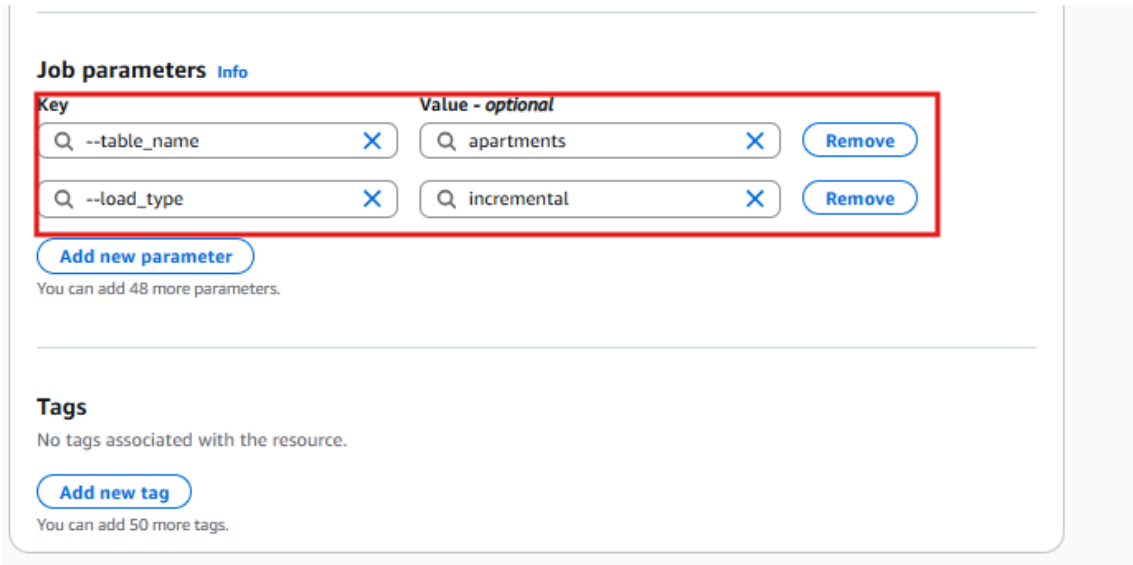
- Timeout → 10 minutes



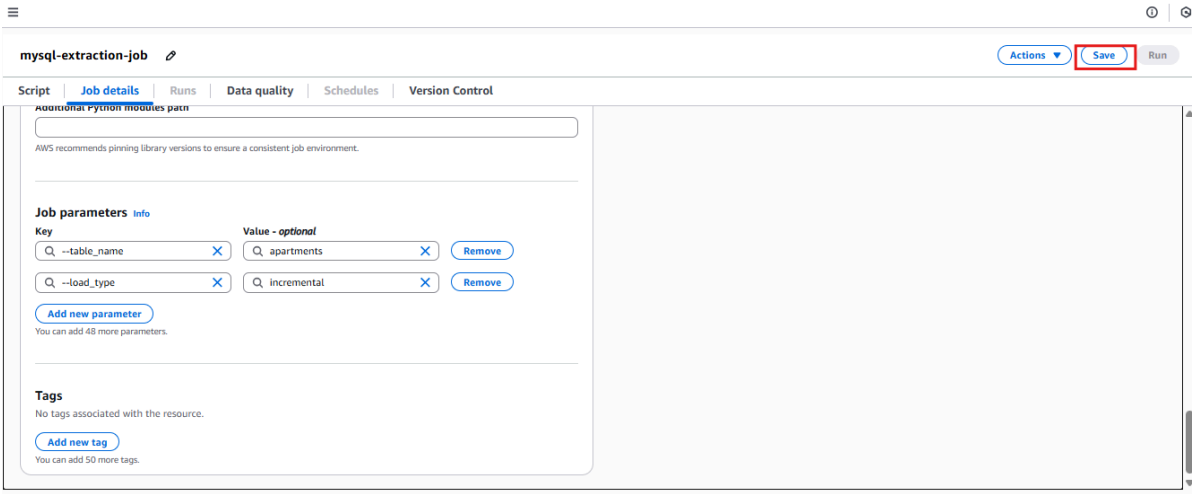
  - Max concurrency → 3



8. Click on Add New Parameter, Under Job Parameters:
   - `-table_name` → apartments
   - `-load_type` → incremental

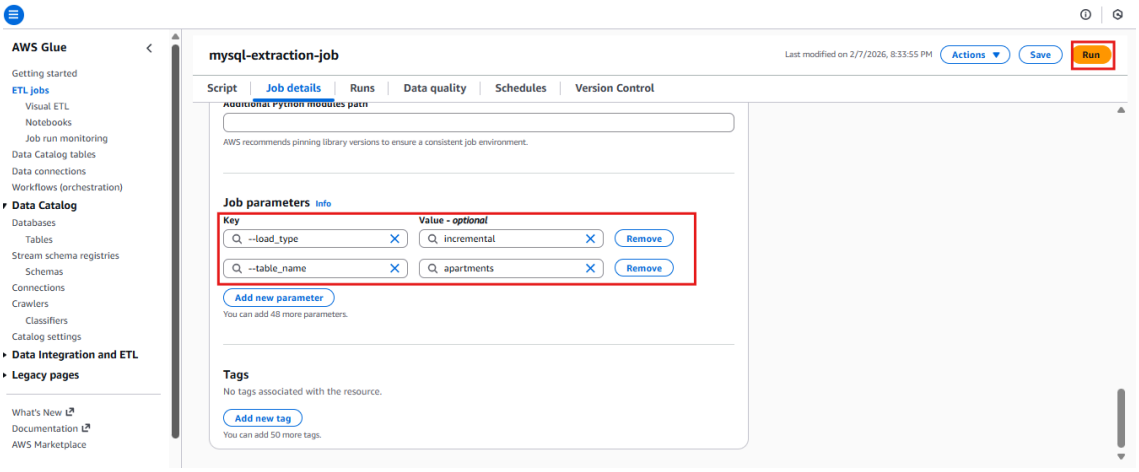


9. Click **Save**.



Step 6: Execute Job for Incremental Tables

1. Run job for:
   - `apartments`
   - `apartment_viewings`



2. Monitor job status in Glue console.
3. Once completed, verify S3 output path:

```
s3://<bucket>/landing-zone/apartment-db/apartments/data.csv
```

```
s3://<bucket>/landing-zone/apartment-db/apartment_viewing
s/data.csv
```

4. Verify DynamoDB table:
   - `last_extracted_value` should now be updated for:
     - apartments
     - apartment_viewings

Step 7: Execute Job for Full Load Table

1. Modify job parameters:
   - `-table_name` → apartment_attributes
   - `-load` → full (or any non-incremental value)
2. Save and run job.
3. Verify output in S3:

```
s3://<bucket>/landing-zone/apartment-db/apartment_attribut
es/data.csv
```

4. DynamoDB:
   - load column remains NULL
   - no incremental tracking required

Outcome

A Glue Python Shell job was successfully configured to extract data from Amazon Aurora MySQL.

Incremental loads were managed using DynamoDB configuration tracking.

Extracted data was written to S3 landing zone, and incremental metadata was updated correctly.

The data is now ready for downstream ingestion into Amazon Redshift in the next pipeline stage.